

KHATANGA, Russian Federation

WMO: 208910

Lat: **71.977N**Lon: **102.466E**Elev: **108**StdP: **14.64**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-52.8	-48.9	-56.3	0.2	-52.6	-52.6	0.2	-48.5	31.7	1.5	28.3	-1.3	5.2	50	0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	75.8	61.9	71.4	59.5	66.9	57.3	63.5	73.7	60.7	70.0	57.8	65.7	9.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	74.9	68.4	56.0	67.0	65.0	53.4	60.9	62.8	29.0	73.9	27.0	70.1	25.1	65.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	21.9	18.6	-56.3	83.5	4.8	4.2	-59.7	86.6	-62.5	89.1	-65.2	91.4	-68.7	94.5		
(5)			-56.3	67.2	4.8	2.6	-59.7	69.1	-62.5	70.6	-65.2	72.1	-68.7	74.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-23.2	-25.6	-13.5	3.8	22.6	45.2	55.2	49.3	36.0	11.9	-10.5	-20.6		
	DBStd	31.50	16.32	14.99	16.91	13.32	10.51	10.36	8.21	6.98	8.05	13.35	13.87	15.28		
	HDD50	14550	2268	2117	1968	1387	850	214	37	101	424	1181	1814	2189		
	HDD65	19701	2733	2537	2433	1837	1315	603	322	488	869	1646	2264	2654		
	CDD50	352	0	0	0	0	1	70	198	79	4	0	0	0		
	CDD65	27	0	0	0	0	0	8	18	1	0	0	0	0		
	CDH74	203	0	0	0	0	0	73	124	6	0	0	0	0		
	CDH80	37	0	0	0	0	0	18	19	0	0	0	0	0		
	WSAvg	9.5	9.6	8.9	9.3	9.9	10.4	10.5	9.7	9.5	9.8	9.3	8.8	8.7		
	PrecAvg	11.1	0.6	0.5	0.6	0.6	0.7	1.3	1.6	1.5	1.3	1.1	0.8	0.7		
(15) Precipitation	PrecMax	16.4	1.6	1.3	2.1	1.7	1.4	4.5	4.4	3.7	3.3	2.6	2.0	3.6		
	PrecMin	6.2	0.0	0.1	0.1	0.0	0.1	0.0	0.2	0.2	0.0	0.4	0.2	0.0		
	PrecStd	2.2	0.4	0.3	0.4	0.4	0.4	1.0	0.8	0.9	0.7	0.5	0.4	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	20.9	30.7	42.8	53.4	82.4	82.8	75.3	62.6	41.4	27.2	18.7		
		MCWB	18.4	20.2	28.8	37.1	45.5	63.4	65.5	61.3	52.1	38.2	26.0	17.6		
	2%	DB	12.4	11.9	25.1	37.5	46.0	74.1	77.4	70.2	55.4	36.1	19.2	13.9		
		MCWB	11.6	11.3	23.7	33.7	39.6	60.3	63.1	58.5	48.4	34.5	18.5	13.0		
	5%	DB	6.9	4.6	19.7	32.0	41.3	68.0	73.4	65.5	50.1	33.5	14.5	8.8		
		MCWB	6.2	4.0	18.3	29.5	36.6	56.7	61.1	56.9	45.6	32.5	13.7	8.0		
	10%	DB	1.1	-2.7	13.5	25.0	37.2	61.2	69.3	61.0	46.4	30.3	9.4	2.8		
		MCWB	0.6	-3.1	12.4	23.0	34.0	52.4	59.1	54.4	43.2	28.9	8.7	2.2		
	0.4%	WB	18.4	20.1	29.2	37.6	45.8	64.8	68.2	63.6	53.8	39.2	25.9	17.8		
		MCDB	18.8	20.7	30.9	42.2	52.1	78.9	79.0	70.8	61.0	40.6	27.1	18.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.7	11.2	23.7	33.9	40.1	61.2	64.8	59.9	48.9	34.6	18.3	13.0		
		MCDB	12.3	11.8	25.2	37.0	44.8	73.0	74.2	67.0	53.8	35.7	18.9	13.7		
	5%	WB	6.1	3.9	18.3	29.4	36.8	57.5	62.4	57.5	46.0	32.4	13.7	7.9		
		MCDB	6.8	4.5	19.6	32.1	40.7	67.3	72.1	63.9	49.7	33.4	14.5	8.6		
	10%	WB	0.4	-3.3	12.5	23.0	34.1	52.6	59.6	55.0	43.4	28.7	8.7	2.1		
		MCDB	0.9	-2.9	13.6	24.9	36.7	60.9	68.3	61.2	46.1	30.2	9.4	2.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.1	15.2	17.5	12.3	12.2	12.9	12.1	9.2	10.6	11.9	12.4		
		MCDBR	16.9	16.8	14.7	19.0	15.0	21.1	19.5	19.8	16.1	9.9	16.0	17.2		
		MCWBR	16.7	16.7	13.8	16.3	10.7	11.5	10.2	11.1	10.5	8.8	15.8	16.9		
		MCDBR	17.0	17.1	14.4	19.1	14.5	20.4	18.3	17.1	14.4	9.6	15.8	17.3		
	5% WB	MCWBR	16.8	17.0	13.7	16.6	10.7	12.2	10.8	11.0	10.1	9.1	15.6	17.0		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.203	0.289	0.313	0.343	0.346	0.383	0.381	0.306	0.210	N/A	N/A		
	taud		N/A	1.942	2.010	2.072	2.121	2.423	2.352	2.298	2.410	1.998	N/A	N/A		
	Ebn at Noon		0	133	206	250	258	266	247	228	215	122	0	0		
	Edh at Noon		0	10	25	34	38	30	31	28	18	9	0	0		
(44) All-Sky Solar Radiation	RadAvg		0	69	531	1325	1866	1743	1553	947	436	126	0	0		
	RadStd		0	7	43	68	151	156	136	125	49	19	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +2.05	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page